

Understanding Rates

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A **rate** is a comparison of two quantities
measured in **different units**.

Writing Rates

Rates are often written using the word “per,” as a fraction, or with a colon.

60 kilometers per hour

$$\frac{60 \text{ kilometers}}{1 \text{ hour}}$$

Finding a Rate

To find a rate, divide one quantity by another quantity with a different unit.

$$\text{Rate} = \frac{\text{Quantity 1}}{\text{Quantity 2}}$$

Unit Rate

A **unit rate** is a rate with a denominator of 1 unit.

For example:

$$\frac{\text{P}240}{6 \text{ kg}} = \frac{\text{P}40}{1 \text{ kg}}$$

The unit rate is P40 per kilogram.

Examples

Example 1

A bicycle travels 48 km in 4 hours. Find the rate in kilometers per hour.

$$\text{Rate} = \frac{48}{4} = 12 \text{ km/h}$$

Example 2

A worker earns ₱540 for 9 hours of work. Find the rate per hour.

$$\text{Rate} = \frac{\cancel{\text{₱}540}}{9} = \cancel{\text{₱}}60 \text{ per hour}$$

Example 3

A grocery store sells 8 kg of sugar for ₱480.
Find the rate per kilogram.

$$\text{Rate} = \frac{\text{₱}480}{8} = \text{₱}60 \text{ per kg}$$

Practice Exercises

Practice 1

A faucet fills 36 liters of water in 6 minutes.
Find the rate in liters per minute.

Practice 2

A typist types 420 words in 7 minutes. Find the rate in words per minute.

Practice 3

A bus travels 315 km in 5 hours. Find the rate in kilometers per hour.

Practice 4

A machine produces 720 bottles in 8 hours.
Find the rate in bottles per hour.

Practice 5

A student reads 90 pages in 3 hours. Find the rate in pages per hour.

Practice 6

A farmer harvests 150 kilograms of vegetables in 5 days. Find the rate in kilograms per day.

Practice 7

A baker uses 24 cups of flour to make 12 batches of bread. Find the rate in cups per batch.

Activities

Activity 1

A bicycle travels 60 km in 5 hours. Find the rate in kilometers per hour.

Activity 2

A worker earns ₱700 for 10 hours of work. Find the rate per hour.

Activity 3

A grocery store sells 6 kg of sugar for ₱390.
Find the rate per kilogram.

Activity 4

A faucet fills 45 liters of water in 5 minutes.
Find the rate in liters per minute.

Activity 5

A typist types 540 words in 9 minutes. Find the rate in words per minute.

Activity 6

A bus travels 420 km in 6 hours. Find the rate in kilometers per hour.

Activity 7

A machine produces 900 bottles in 9 hours.
Find the rate in bottles per hour.

Activity 8

A student reads 120 pages in 4 hours. Find the rate in pages per hour.

Activity 9

A farmer harvests 210 kilograms of vegetables in 7 days. Find the rate in kilograms per day.

Activity 10

A baker uses 30 cups of flour to make 15 batches of bread. Find the rate in cups per batch.

Thank you!